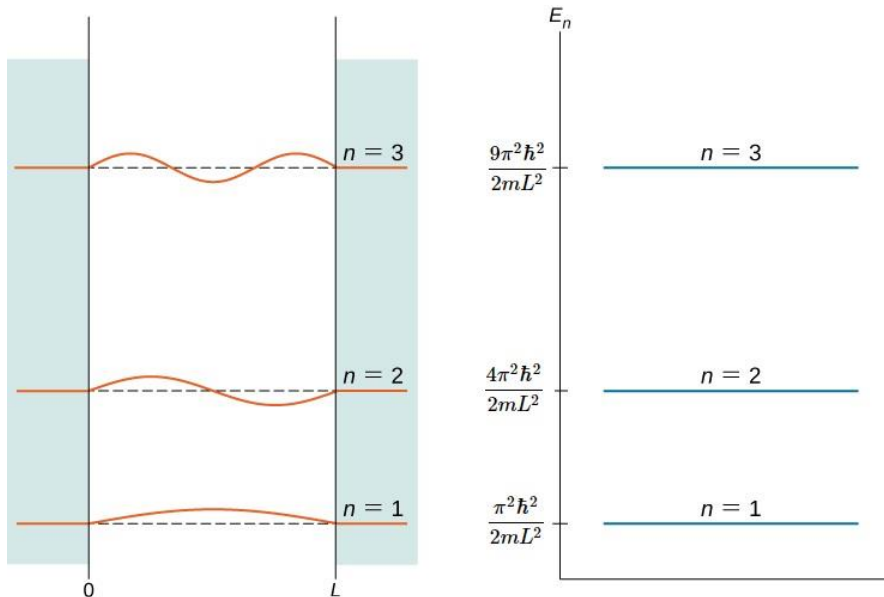
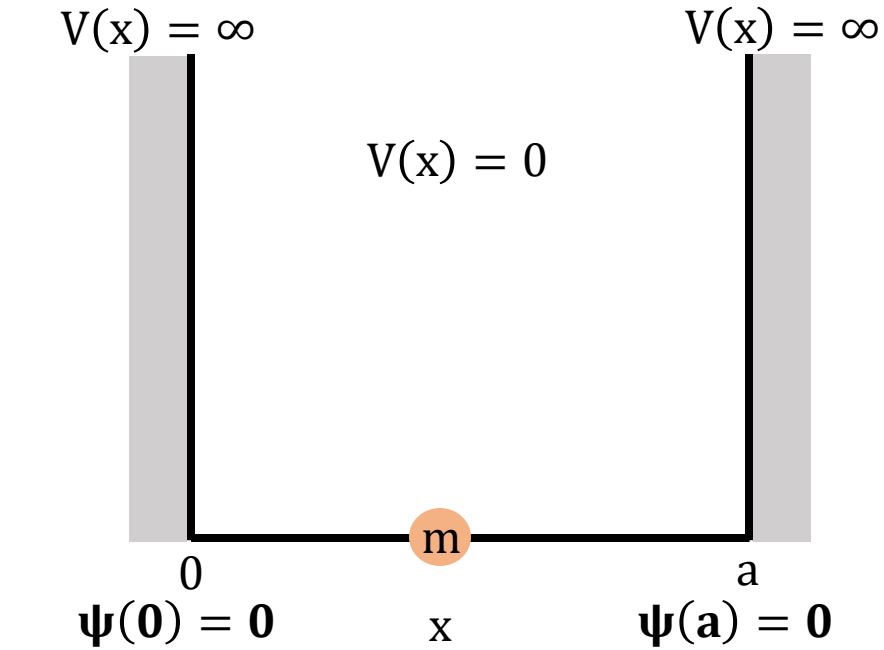


Particle in a Box Continued and Tunneling

1. Calculate the expectation values of observables in the particle in a box system.
2. Explain the physical significance of the expectation values of position and momentum for the particle in a box.
3. Show that the spatial wavefunctions found in the time-independent Schrödinger equation are stationary states.
4. Qualitatively describe the behavior of a quantum mechanical particle for several 1D potential systems.
5. Identify real-world examples of quantum mechanical tunneling.
6. Write quantum mechanical problems in bra-ket notation.



SE

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) = E\psi(x)$$

Solve

Wavefunctions

$$\psi(x) = \begin{cases} 0, & 0 > x > a \\ \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right), & 0 < x < a \end{cases}$$

Energy

$$E_n = \frac{n^2 \pi^2 \hbar^2}{a^2 2m} = \frac{n^2 h^2}{8ma^2}$$

Observables

Energy

$$E_n = \frac{n^2 \pi^2 \hbar^2}{a^2 2m} = \frac{n^2 \hbar^2}{8ma^2} \quad n = 1, 2, 3, \dots$$

Position

$$\hat{x}\psi_n(x) = x\psi_n(x) \quad \text{Not an eigenfunction}$$

$$\langle x \rangle = \int_0^a \psi_n^* x \psi_n dx \Rightarrow \langle x \rangle = \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) x \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) dx$$

$$\langle x \rangle = \frac{2}{a} \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) x dx = \frac{2}{a} \left(\frac{a^2}{4}\right) = \frac{a}{2} \quad \text{The middle of the box}$$

Momentum

$$\hat{p}\psi_n(x) = -i\hbar \frac{d}{dx} \psi_n(x) \propto \cos\left(\frac{n\pi x}{a}\right) \quad \text{Not an eigenfunction}$$

$$\langle p \rangle = - \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) i\hbar \frac{d}{dx} \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) dx = -\frac{2}{a} i\hbar \int_0^a \sin\left(\frac{n\pi x}{a}\right) \cos\left(\frac{n\pi x}{a}\right) dx = 0$$

Equally likely to go in either direction

Observables

Energy

$$E_n = \frac{n^2 \pi^2 \hbar^2}{a^2 2m} = \frac{n^2 \hbar^2}{8ma^2} \quad n = 1, 2, 3, \dots$$

Position

$$\hat{x}\psi_n(x) = x\psi_n(x) \quad \text{Not an eigenfunction}$$

$$\langle x \rangle = \int_0^a \psi_n^* x \psi_n dx \Rightarrow \langle x \rangle = \int_0^a \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) x \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) dx$$

$$\langle x \rangle = \frac{2}{a} \int_0^a \sin^2\left(\frac{n\pi x}{a}\right) x dx = \frac{2}{a} \left(\frac{a^2}{4}\right) = \frac{a}{2} \quad \text{The middle of the box}$$

Momentum

$$\hat{p}^2 \psi_n(x) = -\hbar^2 \frac{d^2}{dx^2} \psi_n(x) \propto \sin\left(\frac{n\pi x}{a}\right) \quad \text{Is an eigenfunction}$$

$$\hat{p}^2 \psi_n(x) = -\hbar^2 \frac{d^2}{dx^2} \sin\left(\frac{n\pi x}{a}\right) = \hbar^2 \frac{n^2 \pi^2}{a^2} \sin\left(\frac{n\pi x}{a}\right) = \frac{n^2 \hbar^2}{4a^2} \psi(x)$$

$$p^2 = \frac{n^2 \hbar^2}{4a^2} \Rightarrow p = \pm \frac{n\hbar}{2a}$$

Observables

Position

$$\langle x \rangle = \frac{2}{a} \int_0^a \sin^2 \left(\frac{n\pi x}{a} \right) x dx = \frac{2}{a} \left(\frac{a^2}{4} \right) = \frac{a}{2}$$

Momentum

$$p^2 = \frac{n^2 h^2}{4a^2} \Rightarrow \mathbf{p} = \pm \frac{nh}{2a}$$

Heisenberg Uncertainty Principle

$$\Delta p_x \approx \frac{nh}{2a} - \left(-\frac{nh}{2a} \right) = 2 \left(\frac{nh}{2a} \right) = \frac{nh}{a}$$

Particle can be
anywhere in the box

$$\Delta x \approx a$$

Particle can move in
either direction

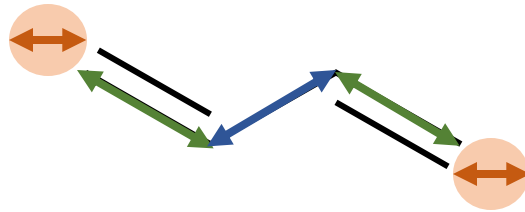
$$\Delta p_x \Delta x \approx \frac{nh}{a} (a) = nh \geq \frac{\hbar}{2}$$

What happens if we just decrease the size of the box?

$$\Delta p_x \approx \frac{nh}{\mathbf{a}}$$

← The uncertainty in the
momentum increases

Applications



Conjugated Butadiene

Delocalized pi electrons across a linear molecule

What is the length of the “box” in this case?

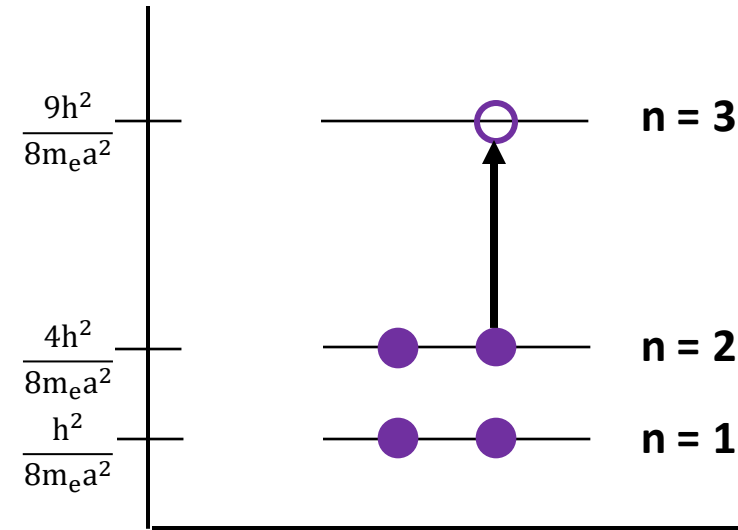
C – C bond length = 154 pm

C = C bond length = 135 pm

C atom radius = 77.0 pm

$$a \approx 2 \times 135 \text{ pm} + 154 \text{ pm} + 2 \times 77.0 \text{ pm}$$

$$\approx 578 \text{ pm}$$



$$\Delta E = \frac{h^2}{8m_e a^2} (3^2 - 2^2)$$

$$\Delta E = \frac{(6.626 \times 10^{-34} \text{ J} \cdot \text{s})^2 5}{8(9.109 \times 10^{-31} \text{ kg})(578 \times 10^{-12} \text{ m})^2} = 9.02 \times 10^{-19} \text{ J}$$

$$\tilde{\nu} = \frac{\Delta E}{hc} = 4.54 \times 10^4 \text{ cm}^{-1}$$

Experimental: **$4.61 \times 10^4 \text{ cm}^{-1}$**

What about time-dependence?

Find time-dependent wavefunction factor

Time
independent SE

$$\hat{H}\Psi(\mathbf{x}, t) = i\hbar \frac{\partial \Psi(\mathbf{x}, t)}{\partial t}$$

Found by
separation of
variables

$$i\hbar \frac{1}{\chi(t)} \frac{d}{dt} \chi(t) = E$$

$$\frac{1}{\chi} d\chi = E i \hbar dt$$

Integrate

$$\int \frac{1}{\chi} d\chi = \int -\frac{i}{\hbar} E dt$$

Rearrange

$$\ln(\chi) = -\frac{i}{\hbar} Et + C$$

$$\chi(t) = \chi(0)e^{-iEt/\hbar}$$

Find square of total wavefunction

$$\Psi(\mathbf{x}, t) = \psi(\mathbf{x})\chi(t)$$

$$\Psi(\mathbf{x}, t) = \psi(\mathbf{x})e^{-iEt/\hbar}$$

$$\Psi_n(\mathbf{x}, t) = \psi_n(\mathbf{x})e^{-iE_n t/\hbar}$$

$$\Psi_n^*(\mathbf{x}, t)\Psi_n(\mathbf{x}, t)dx = \psi_n^*(\mathbf{x})e^{\frac{iE_n t}{\hbar}}\psi_n(\mathbf{x})e^{-\frac{iE_n t}{\hbar}}dx$$

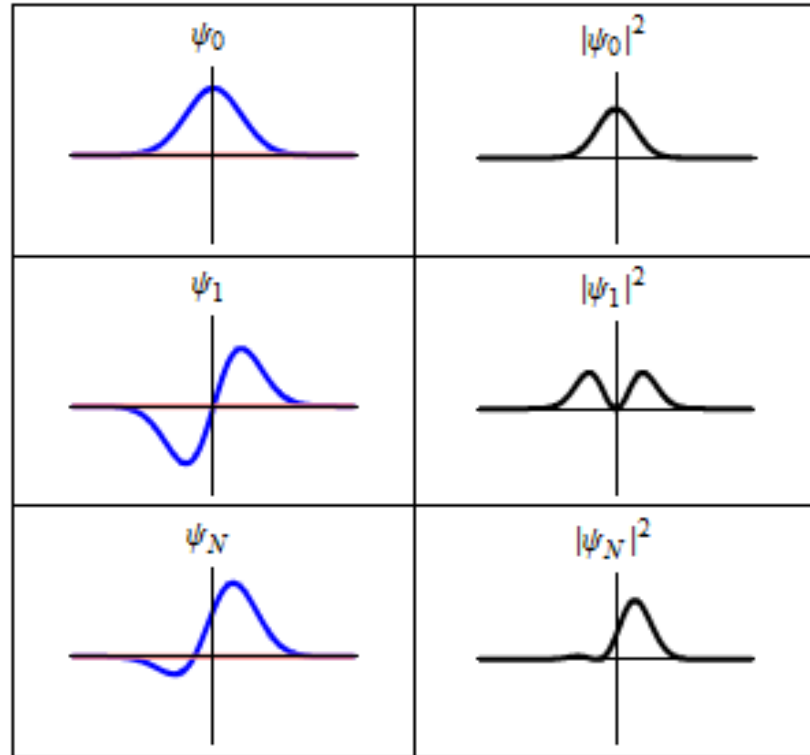
$$\Psi_n^*(\mathbf{x}, t)\Psi_n(\mathbf{x}, t)dx = \psi_n^*(\mathbf{x})\psi_n(\mathbf{x})dx$$

$\psi_n(\mathbf{x})$ are **stationary states**

Observable properties of the wavefunction are
constant in time

Similar to **orbitals** in atoms and molecules!

What about time-dependence?



You can determine the time-dependent behavior as a **superposition** of the **stationary states**!

Find square of total wavefunction

$$\Psi(\mathbf{x}, t) = \psi(\mathbf{x})\chi(t)$$

$$\Psi(\mathbf{x}, t) = \psi(\mathbf{x})e^{-iEt/\hbar}$$

$$\Psi_n(\mathbf{x}, t) = \psi_n(\mathbf{x})e^{-iE_nt/\hbar}$$

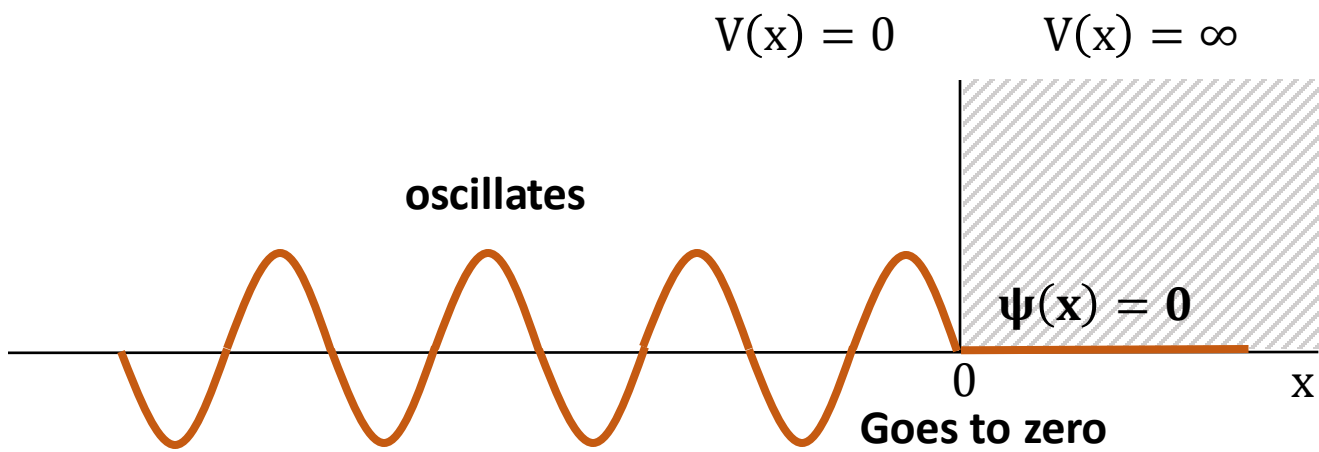
$$\Psi_n^*(\mathbf{x}, t)\Psi_n(\mathbf{x}, t)dx = \psi_n^*(\mathbf{x})e^{\frac{iE_nt}{\hbar}}\psi_n(\mathbf{x})e^{-\frac{iE_nt}{\hbar}}dx$$

$$\Psi_n^*(\mathbf{x}, t)\Psi_n(\mathbf{x}, t)dx = \psi_n^*(\mathbf{x})\psi_n(\mathbf{x})dx$$

$\psi_n(\mathbf{x})$ are **stationary states**

Observable properties of the wavefunction are **constant** in time

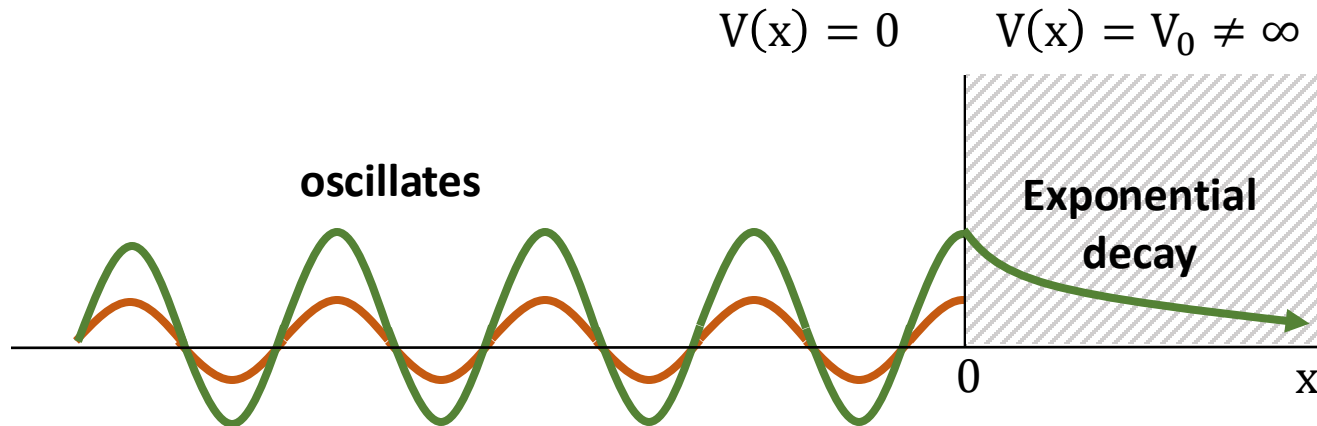
Similar to **orbitals** in atoms and molecules!



Reflects back with same
energy (λ) and amplitude

$$\psi(x) = c_1 e^{k_1 i x} + c_2 e^{-k_1 i x} \quad \boxed{k_1^2 = \frac{2mE}{\hbar^2}}$$

~~$$\psi(x) = c_3 \cos(k_1 x) + c_4 \sin(k_1 x)$$~~



Reflects back with same
energy (λ) but with smaller
amplitude

Left of the barrier

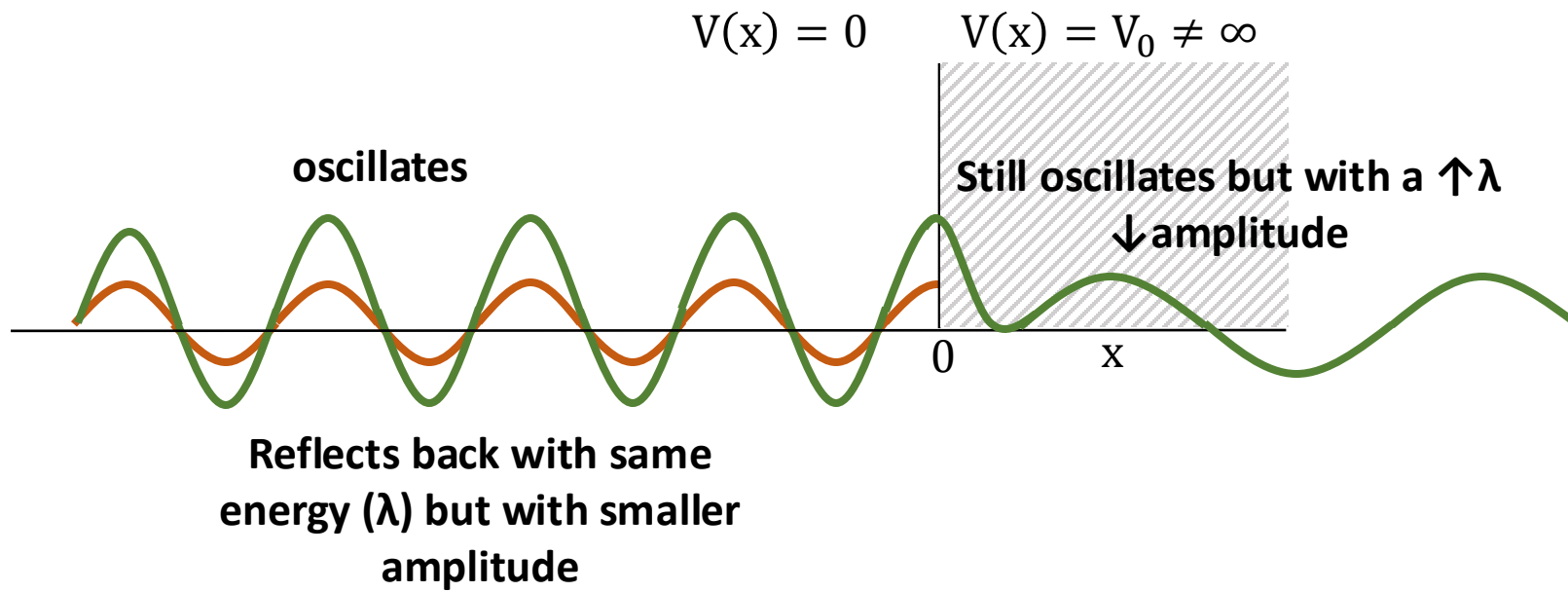
$$\psi(x) = c_1 e^{k_1 i x} + c_2 e^{-k_1 i x}$$

Inside the barrier

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + \underbrace{(V_0 - E)}_{> 0} \psi(x) = 0$$

~~$$\psi(x) = A e^{k_2 x} + B e^{-k_2 x}$$~~

$$\boxed{k_2^2 = \frac{2m(V_0 - E)}{\hbar^2}}$$



Left of the barrier

$$\psi(x) = c_1 e^{k_1 i x} + c_2 e^{-k_1 i x}$$

Inside the barrier

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + \underbrace{(V_0 - E)}_{< 0} \psi(x) = 0$$

$$\psi(x) = c_3 e^{k_1 i x} + \cancel{c_4 e^{-k_1 i x}}$$

Left of the barrier

$$\psi(x) = c_1 e^{k_1 i x} + c_2 e^{-k_1 i x}$$

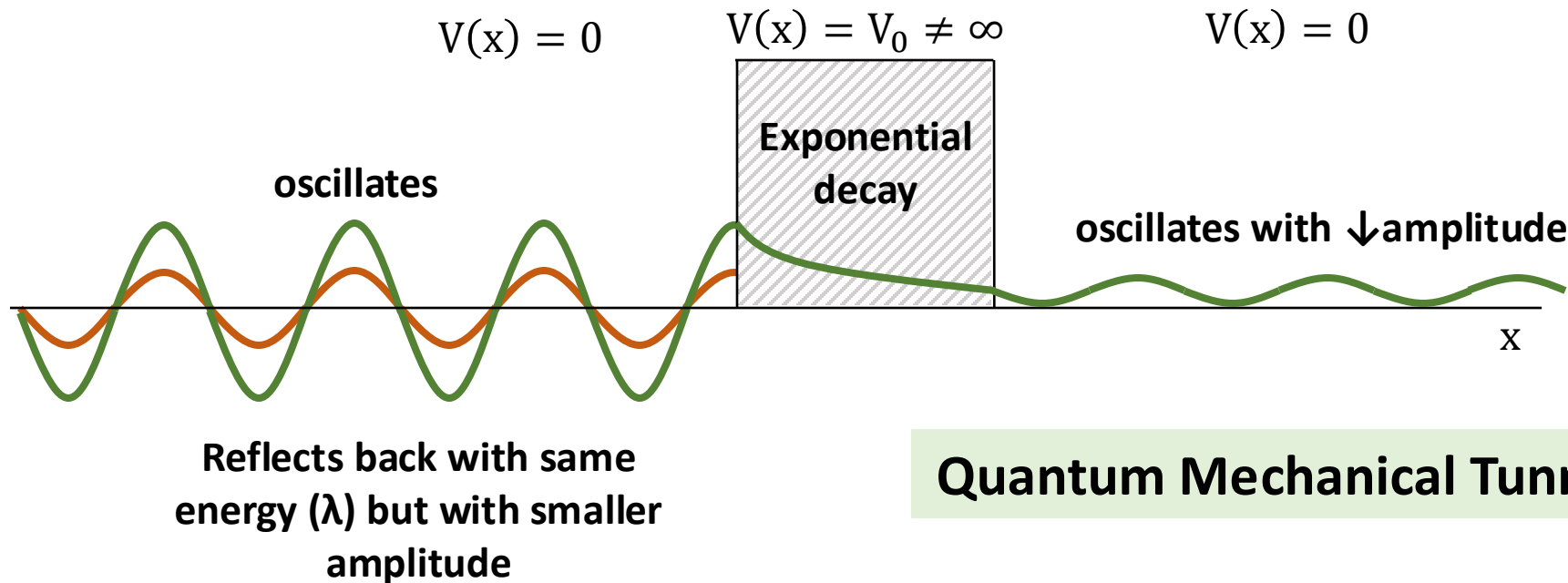
Inside the barrier

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + \underbrace{(V_0 - E)}_{> 0} \psi(x) = 0$$

$$\psi(x) = B e^{-k_2 x}$$

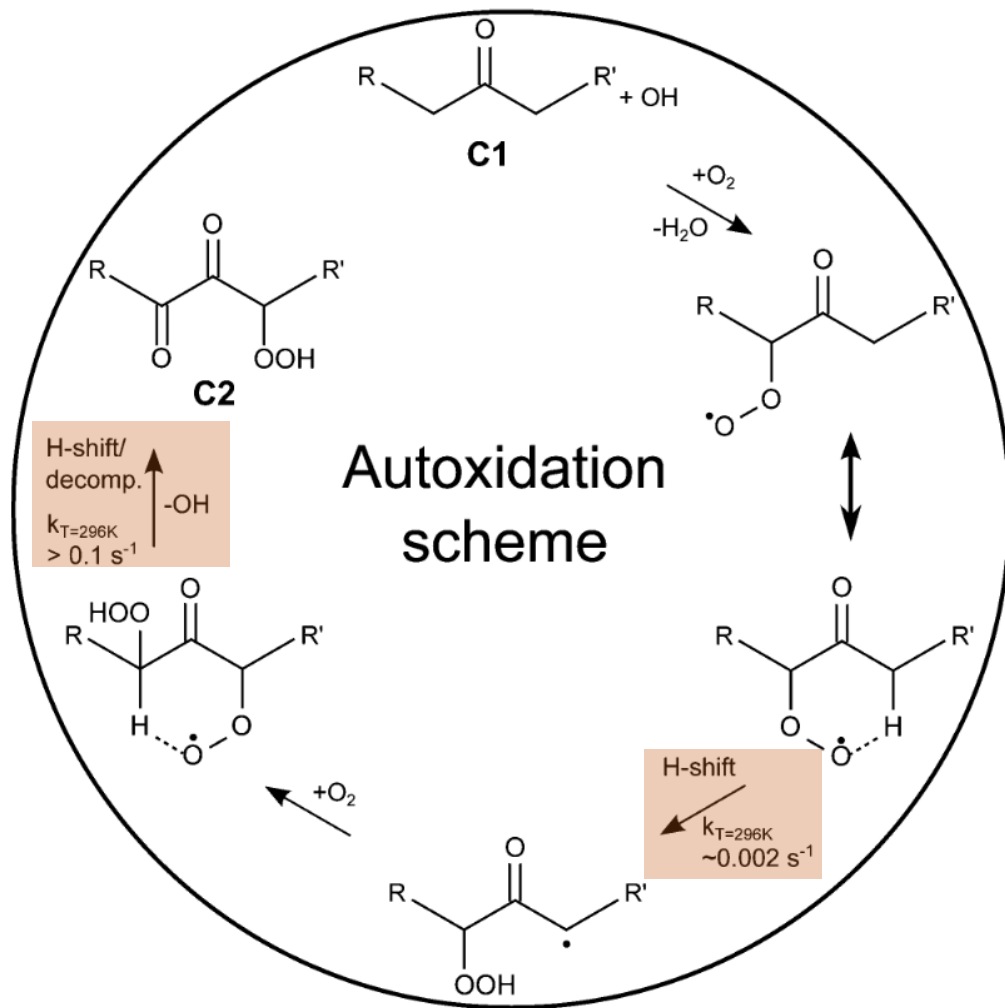
Right of the barrier

$$\psi(x) = c_1 e^{k_1 i x}$$



Quantum Mechanical Tunneling

Problem in atmospheric chemistry – why are there so many compounds with high O/C ratios in atmospheric particles?

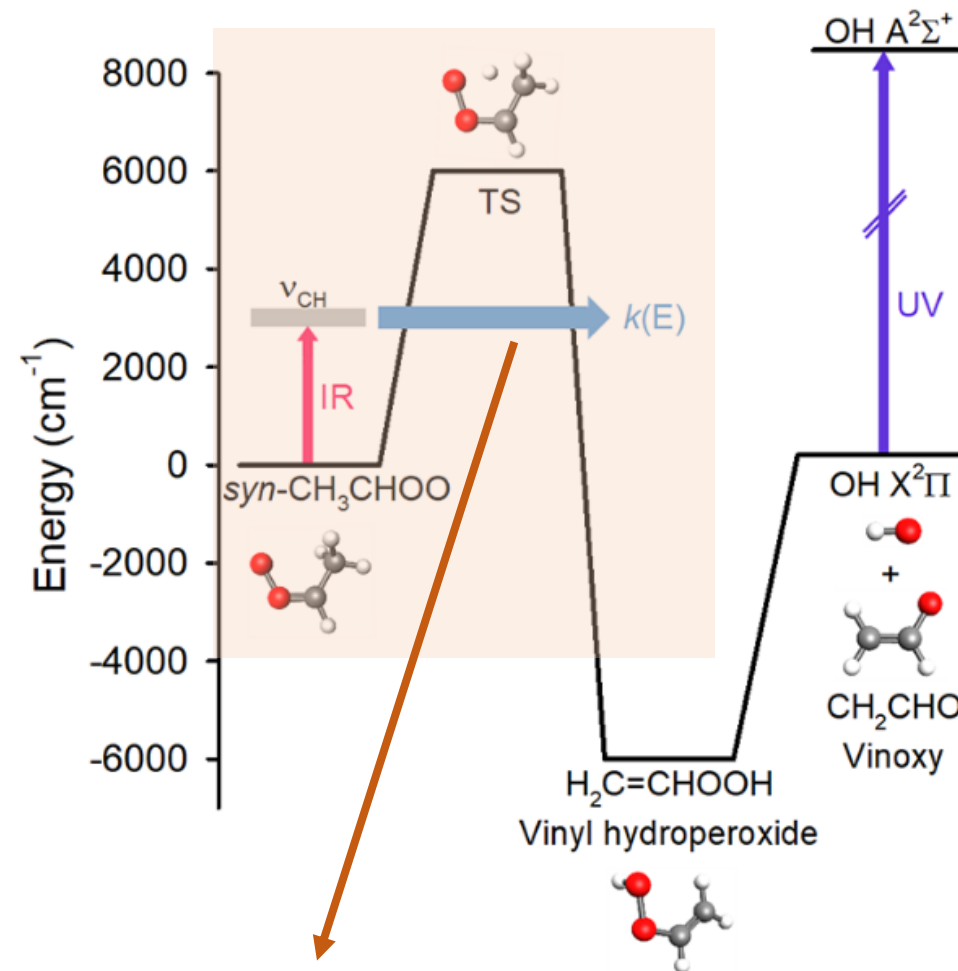


Name	TS	$\tilde{\nu}_{\text{IMAG}}^a$ (cm^{-1})	F^b/R^c (kcal mol^{-1})	κ^d	$k(\rightarrow)^d$ (s^{-1})	$k(\leftarrow)^d$ (s^{-1})	F_{isom}^e
Alkane (2R,4R)		1698	22.49/8.27	27	7.98×10^{-4}	1.75×10^7	0.0088
γOOH (2S,4R)		1698	20.11/-	174	2.27×10^{-1}	-D ^f	0.92
βO (2R,4S)		1946	23.47/19.72	790	6.69×10^{-3}	2.18×10^0	0.25
$\beta\text{O-OOH}$ (2R,4R)		1848	20.39/-	383	2.88×10^{-1}	-D ^f	0.94

Proton Tunneling

Disappearance of proton in one location and reappearance in another

Production of OH by Criegee intermediates



Tunneling through the transition state barrier!

Particle in a finite potential well

Inside the well

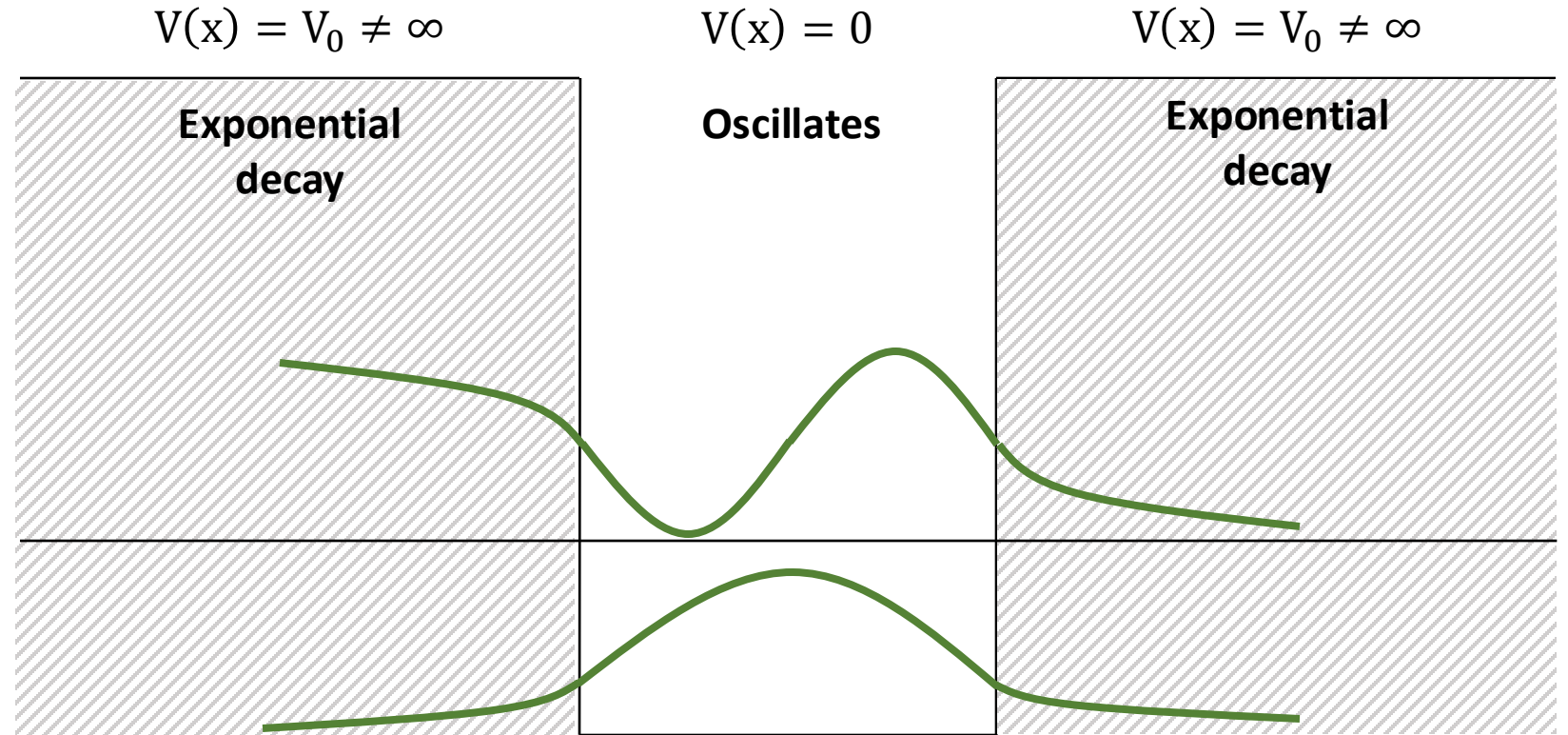
$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) - E\psi(x) = 0$$

$$\psi(x) = c_1 e^{k_1 i x} + c_2 e^{-k_1 i x}$$

Outside the well

$$-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi(x) + \underbrace{(V_0 - E)}_{> 0} \psi(x) = 0$$

$$\psi(x) = A e^{k_2 x} + B e^{-k_2 x}$$





Bra Ket Notation

$$\int_{\text{a.s.}} \psi_i^* \hat{O} \psi_j d\tau \equiv \langle \psi_i | \hat{O} | \psi_j \rangle$$

Bra

$$\langle \psi_i | = \psi_i^*$$

Ket

$$| \psi_j \rangle = \psi_j$$

Normalization Condition

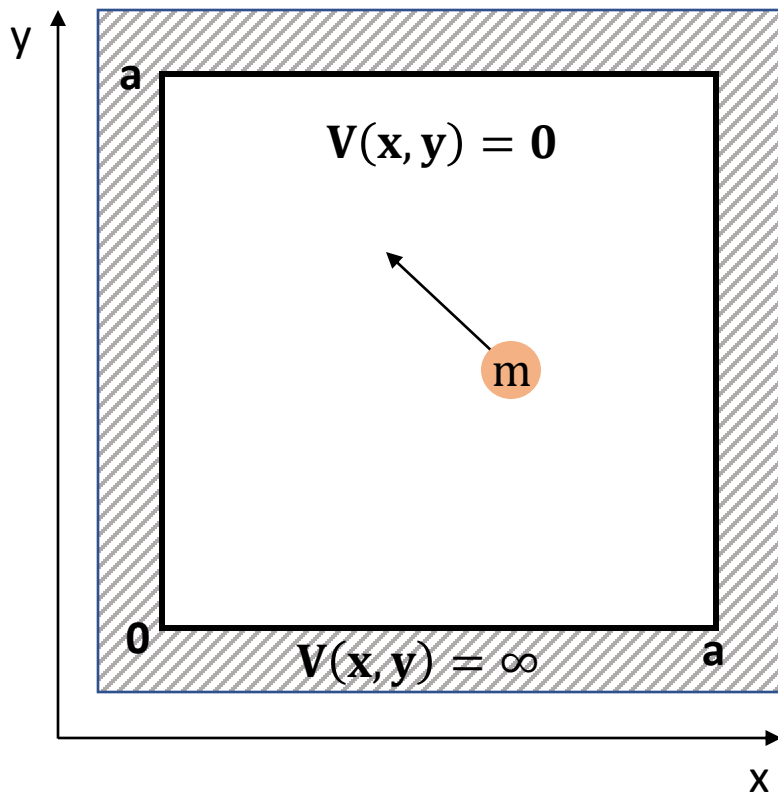
$$\int_{\text{a.s.}} \psi_i^* \psi_i d\tau = 1 \longrightarrow \langle \psi_i | \psi_i \rangle = 1$$

Orthogonality

$$\int_{\text{a.s.}} \psi_i^* \psi_j d\tau = 0 \longrightarrow \langle \psi_i | \psi_j \rangle = 0$$

Definition of a Hermitian Operator

$$\int_{\text{a.s.}} \psi_i^* \hat{O} \psi_j d\tau = \int_{\text{a.s.}} \psi_j \hat{O}^* \psi_i^* d\tau \longrightarrow \langle \psi_i | \hat{O} | \psi_j \rangle = \langle \psi_j | \hat{O} | \psi_i \rangle^*$$



What is the Hamiltonian for this system?

$$H = T + V = \frac{p_x^2 + p_y^2}{2m} + V(x, y) = \frac{p_x^2 + p_y^2}{2m}$$

$$\hat{H} = -\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right)$$

What is the Schrödinger equation for this system?

$$-\frac{\hbar^2}{2m} \left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} \right) \psi(x, y) = E \psi(x, y)$$

Apply separation of variables!

$$\psi(x, y) = X(x)Y(y)$$

$$-\frac{\hbar^2}{2m} \left(Y(y) \frac{\partial^2}{\partial x^2} X(x) + X(x) \frac{\partial^2}{\partial y^2} Y(y) \right) = E X(x)Y(y)$$

$$-\frac{\hbar^2}{2m} \left(\frac{1}{X(x)} \frac{\partial^2}{\partial x^2} X(x) + \frac{1}{Y(y)} \frac{\partial^2}{\partial y^2} Y(y) \right) = E$$

$$-\frac{\hbar^2}{2m} \left(\frac{1}{X(x)} \frac{\partial^2}{\partial x^2} X(x) + \frac{1}{Y(y)} \frac{\partial^2}{\partial y^2} Y(y) \right) = E$$

$$-\frac{\hbar^2}{2m} \frac{1}{X(x)} \frac{\partial^2}{\partial x^2} X(x) = E_x$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial y^2} Y(y) = E_y Y(y)$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} X(x) = E_x X(x)$$

$$-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial y^2} Y(y) = E_y Y(y)$$

These look just like the 1D particle in a box equation

$$X(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_x \pi x}{a}\right)$$

$$Y(y) = \sqrt{\frac{2}{a}} \sin\left(\frac{n_y \pi y}{a}\right)$$

$$\psi(x, y) = \frac{2}{a} \sin\left(\frac{n_x \pi x}{a}\right) \sin\left(\frac{n_y \pi y}{a}\right)$$

$$n_x = 1, 2, 3, \dots \quad E_x = \frac{n_x^2 \hbar^2}{8ma^2}$$

$$E_y = \frac{n_y^2 \hbar^2}{8ma^2} \quad n_y = 1, 2, 3, \dots$$

$$E = E_x + E_y = (n_x^2 + n_y^2) \frac{\hbar^2}{8ma^2}$$

What happens when $n_x = 1$ and $n_y = 2$?

$$\psi(x, y) = \frac{2}{a} \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{2\pi y}{a}\right)$$

$$E = E_x + E_y = (1 + 4) \frac{h^2}{8ma^2} = 5 \frac{h^2}{8ma^2}$$

What happens when $n_x = 2$ and $n_y = 1$?

$$\psi(x, y) = \frac{2}{a} \sin\left(\frac{2\pi x}{a}\right) \sin\left(\frac{\pi y}{a}\right)$$

$$E = E_x + E_y = (4 + 1) \frac{h^2}{8ma^2} = 5 \frac{h^2}{8ma^2}$$

Same particle, but different states!

DEGENERACY

